

Essential Proteins Discovery from Weighted Protein Interaction Networks*

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Abstract. Identifying essential proteins is important for understanding the minimal requirements for cellular survival and development. Fast growth in the amount of available protein-protein interactions has produced unprecedented opportunities for detecting protein essentiality on network level. A series of centrality measures have been proposed to discover essential proteins based on network topology. However, most of them treat all interactions equally and are sensitive to false positives. In this paper, six standard centrality measures are redefined to be used in weighted network. A new method for weighing protein-protein interactions is proposed based on the combination of logistic regression-based model and function similarity. The experimental results on yeast network show that the weighting method can improve the performance of centrality measures considerably. More essential proteins are discovered by the weighted centrality measures than by the original centrality measures used in unweighted network. Even about 20% improvements are obtained from closeness centrality and subgraph centrality.

Keywords: essential protein, protein interaction network, centrality.

1 Introduction

In the post-genome era, the developments of high-throughput methods, such as yeast-two-hybrid and mass spectrometry, have produced vast amounts of protein-protein interaction data, which make it possible for us to study genes and proteins in network level [1]. The corresponding protein interaction networks provide useful insights into cellular effects and functional associations between proteins[2]. Recently, much attention has been paid to the study of the properties of protein interaction networks, including the global properties

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such as “scale-freeness” [3] and “small-world behavior” [4], modularity [5,6] and disassortativity [7].

The analysis of protein interaction networks is to discover the interrelationship between the topological properties and the biological characteristics. An intriguing question is whether the essentiality of a protein can be explained by its placement in the network. A protein is said to be essential for an organism if a knock-out results in lethality or infertility, i.e., the organism cannot survive without it [8,9]. It has been observed in several species, such as *Saccharomyces cerevisiae*, *Caenorhabditis elegans*, and *Drosophila melanogaster* [10,11,12], that proteins with high degree (or hubs) in the network are more likely to be essential than those selected by chance [13]. From the topological perspective, the removal of such high-degree nodes makes the network collapse into isolated clusters. From a biological view, a few highly connected proteins generally guarantee the functional integrity of the network. This phenomenon is commonly referred to as the centrality-lethality rule, which was first observed by Jeong and colleagues [13]. The centrality-lethality rule demonstrates a high correlation between a node’s topological prominence in a protein interaction network and its essentiality. Since then, much attention has been given to the study of high-degree nodes or hubs in protein interaction networks [11,14,15,16,17,18,19]. Most of the authors confirmed the correlation between degree centrality and protein essentiality [11,14,18,19] and some authors examined the reasons for such a correlation [16,17].

Several other topological properties of nodes, such as betweenness centrality [20,21], closeness centrality [22], subgraph centrality [23], eigenvector centrality [24], and information centrality [25], have also been proposed for the discovery of essential proteins, besides the degree centrality. The use of centrality measures based on network topology has become an important means in the study of essential proteins, which is fundamental in many application [26]. For example, the essential genes in pathogenic organisms can be taken as the potential targets for new antibiotics [27] and the identification of essential genes and non-essential genes is valuable for rational drug design [28].

The current centrality measures treat all edges in the network equally. However, some protein-protein interactions are more important than others in reality [16]. Specially, the protein interaction data generated by high-throughput technologies include high false positives [29,30]. Intuitively, the false positives and the real physical interactions can not be treated equally. In this paper, we propose a new method for evaluating the confidence of each interaction and redefined six standard centrality measures in weighted network. The experimental results show that our consideration is particularly meaningful in the discovery of essential proteins. More essential proteins are detected by the weighted centrality measures than by the original centrality measures used in unweighted network.

2 Centrality Measures and Evaluation

A protein interaction network is represented as an undirected graph $G(V, E)$ with proteins as nodes and interactions as edges. Recently, much attention has

been given for the relationship between the centrality of a protein within an interaction network and its essentiality for the organism's survival. There are several commonly used centrality measures for predicting a protein's essentiality, such as degree centrality (DC)[13], betweenness centrality (BC)[20,21], closeness centrality (CC)[22], subgraph centrality(SC)[23], eigenvector centrality(EC)[24], and information centrality(IC)[25]. These different centrality measures have been described in recent reviews [19,31,32].

Though previous authors have tested these centrality measures in protein interaction networks, they did not take the universal false positives into consideration. However, the protein-protein interaction data obtained from large scale, high-throughput experiments generally contain false positives [29,30]. To evaluate how the false positives affect the discovery of essential proteins, we reassessed systematically the six centrality measures (DC, BC, CC, SC, EC, and IC) by using three data sets with different confidence levels. The three test data sets were from von Mering *et al.* 2002 [29]. To describe simply, we name the three test data Y2k (2455 interactions), Y11k (11000 interactions), and Y45k (45000 interactions), respectively, according to the number of edges included in them. Y2k, which consists of the first 2455 reliable interactions, is a subset of Y11k, and Y11k is a subset of Y45k. A list of essential genes was obtained from the MIPS database[33]. For each centrality measure, we use it to rank all proteins in the network according to their centrality values and select a certain number of the top proteins. Similar to most experiments[19], we also select the top 15%, 20%, and 25% proteins as essential candidates. The *precision*, scored as $TP/(TP+FP)$, is used to validate the essential proteins discovery, where *TP*, true positives, are true essential proteins that are in the discovery and *FP*, false positives, are non-essential proteins that are in the prediction. The *precision* of the six centrality measures for uncovering essential proteins in the three test data is shown in Fig.1.

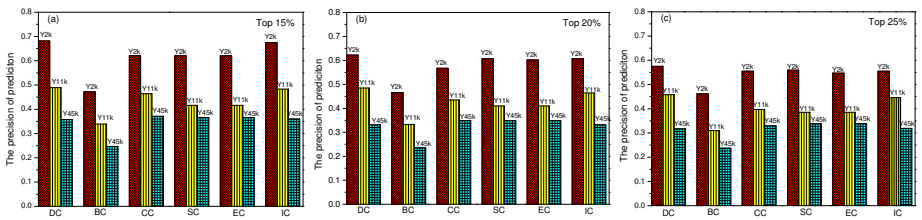


Fig. 1. The precision for essential proteins discovery in three protein interaction networks with different confidence levels. (a)Top 15% proteins are selected, (b)Top 20% proteins are selected, (c)Top 25% proteins are selected.

From Fig.1 we can see that Lower precision is obtained for all centrality measures (DC, BC, CC, SC, EC, and IC) when being used in network of less confidence level to predict the same proportion proteins. As shown in Fig.1 (a), from the high reliable network Y2k to the middle reliable network Y11k, there are

more than 10% decrements. When the unreliable network Y45k is used, all these centrality measures perform on even much less precision. The similar downward trends of precisions for DC, BC, CC, SC, EC, and IC can also be seen in Fig.1 (b) and (c) when the top 20% and 25% proteins are selected. As one can seen in Fig.1, the precision of essential proteins prediction based on network topology depends heavily on the reliability of the network. Therefore, the confidence of the protein-protein interactions should be taken into account when predicting the essential proteins based on network topology. In the following section, we will redefine the six standard centrality measures in weighted protein interaction networks.

3 Definition of Centrality Measures in Weighted Protein Interaction Network

3.1 Definition of Weighted Centrality Measures

A weighted protein interaction network can be represented as a weighted undirected graph $G = (V, E)$. Each edge $(i, j) \in E$ is assigned with a weight $w_{i,j}$, which represents the probability of this interaction between node i and node j being a true positive. To describe simply, the centrality measures (DC [13], BC [20,21], CC [22], SC [23], EC [24], and IC [25]) of a weighted graph G are accordingly marked as DC^W , BC^W , CC^W , SC^W , EC^W , and IC^W .

Definition 1. The weighted degree centrality $DC^W(i)$ of a node i is the sum of weights of the edges connecting node i and its neighbors.

$$DC^W(i) = \sum_{j \in N_i} w_{i,j} \quad (1)$$

where N_i is the set of neighbors of node i .

Definition 2. The weighted betweenness centrality $BC^W(i)$ of a node i is equal to the average fraction of shortest paths that pass through the node i .

$$BC^W(i) = \sum_s \sum_t \frac{\sigma_{st}(i)}{\sigma_{st}}, \quad s \neq t \neq i \quad (2)$$

where σ_{st} denotes the total number of shortest paths between s and t and $\sigma_{st}(i)$ denotes the number of shortest paths from s to t that pass through the node i .

The main difference between $BC^W(i)$ and $BC(i)$ is the calculation of shortest path. In a unweighted graph G , a shortest path between two nodes is a path that has the minimum constituent edges. However, the shorted path may be misdirected with the false positives in protein interaction networks. For example, of all the three paths between node i and node j showed in Fig.2, the path $\{i, v_1, j\}$ will be the choice for the shortest path without regard to the edges' reliability. As a matter of fact, however, the path $\{i, v_4, v_5, v_6, j\}$ is a real pathway from node i to node j . Therefore, we use $c_{i,j} = 1/w_{i,j}$ to describe the cost of edge

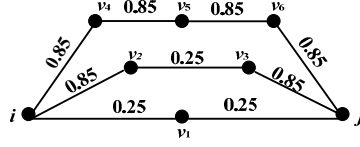


Fig. 2. The shortest path from node i to node j is $\{i, v_4, v_5, v_6, j\}$ with the lowest cost of 4.7

connecting node i to node j . The *cost* of a path is the sum of all the *costs* of its constituent edges. For example, in Fig.2, the *cost* of path $\{i, v_1, j\}$ is 8.0, the *cost* of path $\{i, v_2, v_3, j\}$ is 6.4, the *cost* of path $\{i, v_4, v_5, v_6, j\}$ is 4.7. Obviously, the path $\{i, v_4, v_5, v_6, j\}$ is more likely to be a real pathway with the lowest *cost*. Therefore, the shortest path between a given pair of nodes is the path that has the lowest *cost*. In this paper, we mark the *cost* of a shortest path p from node i to node j as $c_p(i, j)$ for simple and clear description.

Definition 3. The weighted closeness centrality $CC^W(i)$ of a node i in a weighted graph G can be defined as:

$$CC^W(i) = \frac{1}{\sum_{j \neq i} c_p(i, j)} \quad (3)$$

$CC^W(i)$ is a global metric which describes how the given node i connects to other nodes.

Definition 4. The weighted subgraph centrality $SC^W(i)$ of a node i in a weighted graph G can be defined as:

$$SC^W(i) = \sum_{l=0}^{\infty} \frac{\mu_l(i)}{l!} \quad (4)$$

where $\mu_l(i)$ denotes the number of closed walks of length l which starts and ends at node i .

Definition 5. The weighted eigenvector centrality $EC^W(i)$ of a node i in a weighted graph G is defined as the i th component of the principal eigenvector of A , where A is an edge weight matrix. Let λ be an eigenvalue and e be the eigenvector. Then for an equation $\lambda e = Ae$, we can obtain $EC^W(i) = e_1(i)$, where e_1 corresponds to the largest eigenvalue of A .

Definition 6. The weighted information centrality $IC^W(i)$ of a node i in a weighted graph G is defined as:

$$IC^W(i) = \left[\frac{1}{n} \sum_j \frac{1}{I_{ij}} \right]^{-1} \quad (5)$$

where n is the number of nodes in graph G and $I_{ij} = (r_{ii} + r_{jj} - r_{ij})^{-1}$. Let D be a diagonal matrix of the weighted degree of each node and J be a matrix with all its elements equal to one. Then, we get $R = (r_{ij}) = [D - A + J]^{-1}$. For computational purposes, I_{ii} is defined as infinite. Thus, $\frac{1}{I_{ii}} = 0$.

3.2 Construct Weighted Protein Interaction Network

To construct weighted protein interaction network, we assign confidence score to each interaction by combining two aspects: (1) observing its experimental evidences; (2) evaluating its function similarity.

For aspect (1), we use the logistic regression-based model employed in [34,35] to examine the reliability of an interaction. For each interaction, its reliability score is determined by its experimental evidences and the number of observations in each experimental type. The experiments are classed into four categories: co-immunoprecipitation screens, yeast two-hybrid assays, large scale experiments and small scale experiments. The reliability score of an interaction between node i and node j is marked as $L_O(i, j)$ for it is computed based on logistic regression.

For aspect (2), we calculate the function similarity of two connected proteins based on GO (Gene Ontology) semantic similarity. There have been several methods for computing the similarity between GO terms [36,37,38]. Here, we select the widely used method proposed by Lin[37], in which the similarity of two GO terms c_1 and c_2 is defined as:

$$sim(c_1, c_2) = \frac{2 \max_{c \in C_T(c_1, c_2)} (\log p(c))}{\log p(c_1) + \log p(c_2)} \quad (6)$$

where $p(c)$ denotes the probability of encountering term c in the target species and $C_T(c_1, c_2)$ denotes the sets of common ancestors of term c_1 and term c_2 .

Let F_i and F_j denote the sets of function annotations for protein i and protein j , respectively. Then, the function similarity $S_F(i, j)$ of protein i and protein j can be defined as:

$$S_F(i, j) = \max_{c_1 \in F_i, c_2 \in F_j} (sim(c_1, c_2)) \quad (7)$$

To satisfy that the confidence score of each interaction should be between 0 and 1, the following normalization operation is performed.

$$S_F^*(i, j) = \frac{S_F(i, j) - \text{Min_}S_F}{\text{Max_}S_F - \text{Min_}S_F} \quad (8)$$

where $\text{Max_}S_F$ and $\text{Min_}S_F$ denote the maximum value and minimum value of all the interactions' function similarity scores, respectively.

By considering both the reliability measurement and the function similarity, we define the confidence score $C_{score}(i, j)$ of an interaction connecting protein i and protein j as formula (9):

$$C_{score}(i, j) = \frac{L_O(i, j) + S_F^*(i, j)}{2} \quad (9)$$

4 Results

To evaluate whether our method for evaluating the confidences of protein-protein interactions works and to investigate whether the definitions of centrality measures in weighted protein interaction network outperform that in unweighted network, we implement them to predict the essentiality of proteins from *Saccharomyces cerevisiae* for its well characterized by knockout experiments and widely used in previous works[19]. The protein-protein interactions of *Saccharomyces cerevisiae* are downloaded from DIP database[39]. The interaction map is considered as a network in which proteins are represented as nodes and interactions connecting two proteins are represented as edges. There are 4746 proteins and 15166 interactions in total without self-interactions and repeated interactions. For the functional annotation of the proteins, we use functions classified as molecular function by GO[40]. The lethal proteins are obtained from MIPS[33].

Proteins are ranked according to their values of centrality measures and a certain top percentage of proteins are selected as candidates for essential proteins. Then we determine how many of them are essential. For evaluation, we compare the results of centrality measures in weighted network against those in unweighted network. The comparison results are shown in Fig.3.

In Fig.3 we illustrate the number of essential proteins identified by DC^W , BC^W , CC^W , SC^W , EC^W and IC^W as well as by DC , BC , CC , SC , EC and IC with the top 10%, 15%, 20% and 25% of proteins in the protein interaction network. From Fig.3 we can see that all the weighted centrality measures perform significantly better than the unweighted centrality measures in the selection of essential proteins in the yeast protein interaction network. Especially, the improvements of SC^W and CC^W are remarkable for that about 20% extra essential proteins are detected by SC^W and CC^W than by SC and CC . The results also indicate that all the centrality measures based on topological characters are sensitive to the interactions' confidence, which in turn illustrate that effective weighting methods are important.

Take the simplest method DC for example, low-connectivity proteins negatived by it may be essential. On the contrary, non-essential proteins with high interactions may be false predicted. In Fig.4, we give three examples of essential proteins: YDR356W, YNL216W, and YBR060C with 8 interactions. Their weighted degrees are all larger than 4.5. By contrast, the three examples shown in Fig.5 which have more than 20 interactions are validated to be not essential. Their weighted degrees are smaller than 4.0. Of course, we can not say that DC^W can identify all the essential proteins negatived by DC and filtered all the non-essential proteins false predicted by DC . However, more essential proteins are discovered by DC^W than by DC . This phenomenon is more obvious for SC^W and SC , CC^W and CC , BC^W and BC as they are all based on the calculation of path length. And, false positives affect heavily on the calculation of path length as we have discussed previously.

To evaluate all the centrality measures more generally, we cite the jackknifing methodology developed by Holman *et al.*[41]. As shown in Fig.6, proteins are ordered by highest to lowest values of centrality measures and the cumulative count

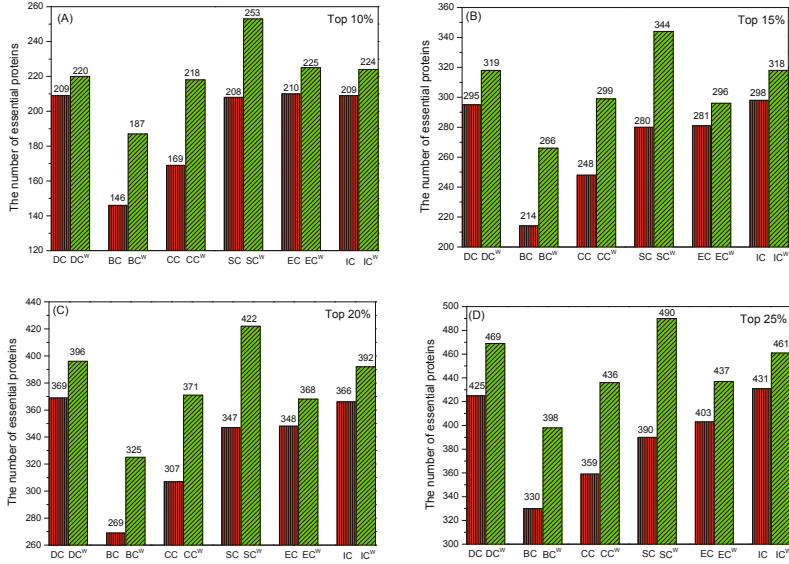


Fig. 3. Comparison of the number of essential proteins that selected from unweighted network and that selected from weighted network by ranking proteins according to their values of centrality measures

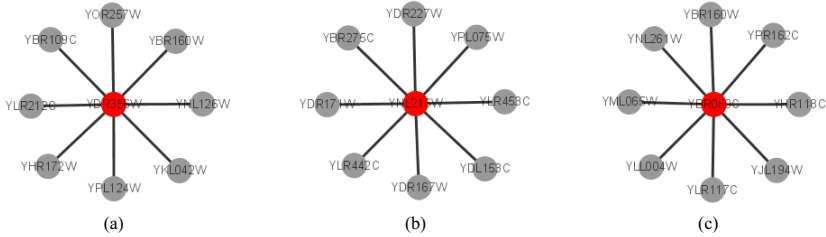


Fig. 4. Examples of low-connectivity essential proteins

of essential proteins is plotted. The area under the curve (AUC) for the weighted centrality measures and that for the unweighted methods are compared. In addition, an ideal ranking is plotted with all essential proteins artificially placed at the beginning of the list. Moreover, 10 random assortments are also plotted for comparison. As can be seen in Fig.6, of all the weighted centrality measures the sorted curves appear to be better than their corresponded unweighted curves. More over, all the sorted curves of centrality measures appear well differentiated from the randomized sorting, which indicates that the discovery of essential proteins based on topological characters are statistical significance.

The above analysis demonstrates that the weighted centrality measures are effective in predicting protein essentiality even false positives existed in the protein

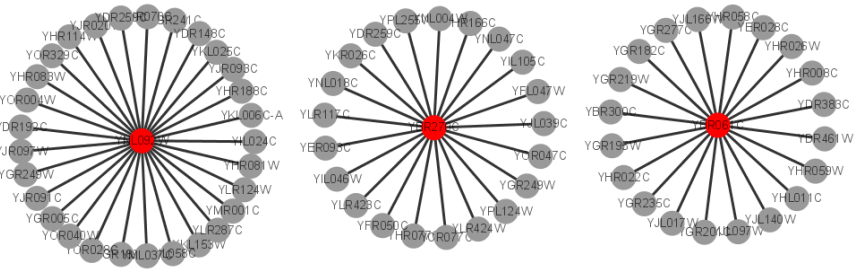


Fig. 5. Examples of high-connectivity non-essential proteins

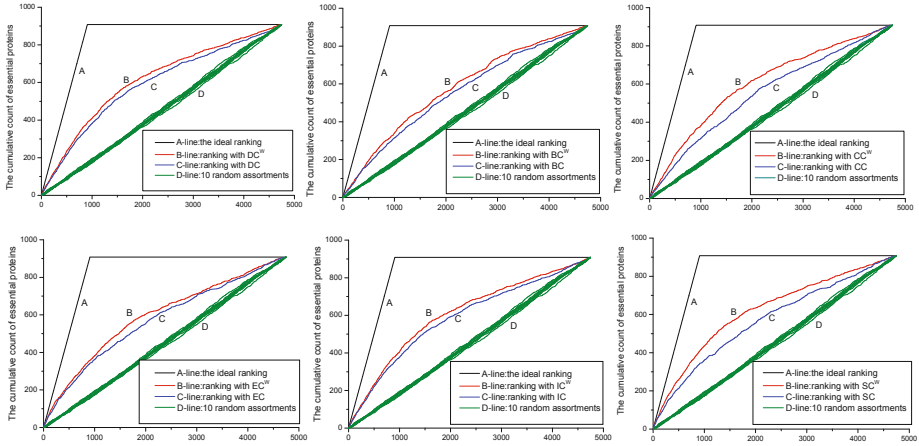


Fig. 6. Essential proteins discovery is validated by a jackknife methodology. The X-axis represents the ranked proteins in the yeast protein interaction network, ranked from left to right as highest to lowest values of centrality measures. The Y-axis is the cumulative count of essential proteins with respect to the ranked proteins moving left to right. Line A is the ideal ranking. Line B is the sorting by weighted centrality measures. Line C is the ranking by unweighted centrality measures. Line D are 10 random assortments.

interaction network. Then, one question is whether the weighted centrality measures are also valid in high confidence protein-protein interactions. We get the top 5283 high reliability interactions from DIP[39] as Core dataset. There are 2373 proteins in total, of which 666 proteins are essential. In Table 1, a comparison of the precision of the weighted centrality measures: DC^W , BC^W , CC^W , SC^W , EC^W and IC^W , and that of the unweighted ones: DC , BC , CC , SC , EC and IC is shown both in the original protein interaction network (All PPIs) and the cleaned network (Core PPIs). The comparison result shows that the weighted centrality measures perform well both in the high positive protein interaction network and the high confidence network.

Table 1. The precision of essential protein discovery on original protein interaction network (All PPIs) and the cleaned network (Core PPIs)

		DC^W	DC	BC^W	BC	CC^W	CC	SC^W	SC	EC^W	EC	IC^W	IC
All PPIs	Top 10%	0.46	0.44	0.39	0.36	0.46	0.36	0.53	0.44	0.47	0.44	0.47	0.44
	Top 15%	0.45	0.42	0.37	0.34	0.42	0.35	0.48	0.39	0.42	0.39	0.45	0.42
	Top 20%	0.42	0.39	0.34	0.32	0.39	0.32	0.44	0.37	0.39	0.37	0.41	0.39
	Top 25%	0.40	0.37	0.34	0.30	0.37	0.30	0.41	0.33	0.37	0.34	0.39	0.36
Core PPIs	Top 10%	0.50	0.50	0.44	0.42	0.42	0.39	0.55	0.52	0.53	0.42	0.47	0.47
	Top 15%	0.49	0.47	0.41	0.40	0.40	0.37	0.50	0.47	0.47	0.39	0.45	0.45
	Top 20%	0.47	0.47	0.41	0.39	0.41	0.36	0.48	0.43	0.44	0.37	0.44	0.41
	Top 25%	0.44	0.43	0.39	0.37	0.40	0.36	0.46	0.41	0.43	0.36	0.42	0.38

5 Conclusions and Future Work

By assessing the performance of six standard centrality measures for the discovery of essential proteins in different confidence-level networks, we find that the centrality measures based on network topology are very sensitive to false positives. Thus, we introduce a new weighing method for evaluating the confidence of protein-protein interactions. The experimental evidences are considered by using the logistic regression-based model. And function similarity between proteins are also considered in our method. Based on the new weighting method, we construct a weighted protein interaction network and redefine the six standard centrality measures in weighted network. The experimental results show that: (1)the weighting method can improve the performance of centrality measures considerably; (2)the weighted centrality measures are significantly to be better than the original centrality measures used in unweighted network, Even about 20% improvements are obtained from closeness centrality and subgraph centrality; (3)the weighted centrality measures perform well both in the high positive networks and high confidence networks; (4)the prediction based on network topology is significantly better than random selection.

The proposed weighing method can also be used in other fields, such as identification of protein complexes and functional modules. As future work, it would be interesting to apply this weighing method to other studies. Moreover, analysis of the relation among different centrality measures and exploitation of an ensemble approach for integrating these different types of prediction methodologies are also our future work.

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